

Enzyme Design Practical: PyRosetta Analysis and Optimization

Data availability

All data for the practical are available in the shared Google Drive directory:

`EnzDesPractical`

This contains the following directories:

- `RFDiff2` – structures generated by RFdiffusion2
- `ligandMPNN` – sequences generated by ligandMPNN
- `AlphaFold3` – structure predictions generated by AlphaFold3
- `ThreadedRFDiff2` – ligandMPNN sequences threaded onto the RFdiffusion2 backbones
- `PyRosetta notebooks` – prepared notebooks for the PyRosetta calculations

Working with PyRosetta

You can approach the PyRosetta calculations in different ways. You can write the notebooks from scratch for additional PyRosetta training, use AI-assisted coding to develop your own code, or use the notebooks we have provided to save time. You can also combine these approaches.

Whichever approach you choose, make sure that you understand what the code is doing. In particular, identify the important variables and options for each calculation and consider how these choices may affect the results.

The goal is not simply to run the calculations, but to understand what is being calculated and how to interpret the results.

Task 1 – Threading

Take the protein design(s) generated by RFdiffusion2 and thread the corresponding ligandMPNN sequence onto the backbone.

Compare the threaded model with the structure predicted by AlphaFold3.

Methods: Identify the main options involved in threading a sequence onto a structure. What is retained from the starting structure, and what is changed?

Discussion:

How different are the threaded and AF3 models? Where do the main differences occur?

What can we learn from comparing a threaded model with an independently predicted structure?

Task 2 – FastRelax

Run FastRelax on both the AF3 and threaded models.

Compare the structures before and after relaxation. Compare the total Rosetta scores of the models where the sequences are identical, and plot their per-residue energies to identify regions with energetic differences.

Examine the active-site residues. Are their positions and orientations maintained during relaxation?

Methods: Identify the main FastRelax options. Which settings are likely to have the greatest effect on the calculation?

Discussion:

How does the structure change during optimization?

What can the total and per-residue energies tell you about the different models?

Compare the threaded, threaded-relaxed, AF3, and AF3-relaxed structures. What evidence would you use to decide how much confidence to place in each model?

Task 3 – Ligand energy analysis

Analyze the interaction between the protein and ligand by calculating the protein–ligand interaction energy.

Examine which residues contribute to the interaction and compare relevant models or designs.

Methods: How is the interaction energy defined and calculated? What options or assumptions are important?

Discussion:

What should we look for as a useful metric of ligand binding?

Is interaction energy sufficient, or should it be combined with structural information about the binding site and catalytic geometry?

Task 4 – Design energetic and structural metrics

Run FastRelax on several different designs and develop a set of metrics that can be used to compare them.

These could include energetic measurements as well as structural properties such as packing, exposed hydrophobic residues, active-site geometry, or other properties that you consider relevant.

Methods: For the metrics you choose, identify what is being measured and which important variables or options affect the calculation.

Discussion:

Why did you choose these metrics?

What does each metric tell you about the quality of a design?

Do the different metrics agree, or do they sometimes give conflicting results?

Task 5 – FastDesign

Use FastDesign to redesign residues around the active site while constraining the catalytic residues.

Compare the resulting sequence and structure with the design generated by ligandMPNN.

Methods: Identify the main FastDesign options. In particular, consider which residues are allowed to change and how the active site is constrained.

Discussion:

What changes did FastDesign make?

How did these changes affect the structure and energy?

How would you compare the FastDesign and ligandMPNN sequences?

Task 6 – $\Delta\Delta G$ prediction

Investigate how the active-site residues contribute to protein stability.

Identify the active-site residues and use CartesianDDG to predict the effect of individually mutating them to alanine.

Methods: Identify the important options in the CartesianDDG calculation. Make sure you understand how the $\Delta\Delta G$ is calculated and how its sign should be interpreted.

Discussion:

Which residues have the largest predicted effect on stability?

Are there signs of a trade-off between the functional requirements of the active site and protein stability?

Do some residues appear to be important for function despite being energetically costly?

Overall discussion

Taken together, these tasks provide different ways of evaluating an enzyme design. Consider what each method tells you, what its limitations are, and how sensitive its results are to the options used.

How would you combine structural, energetic, and active-site information to evaluate a set of enzyme designs?

Bonus Task 7 – Interface analysis

Use the interface analyser in PyRosetta to analyze a protein complex that you have first run FastRelax on.

DiscussionÖ What stabilize the interface between the binding partners?